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MME 365-Engineering Thermodynamics – QUIZ 1

Submission date; Monday, 21st March, 2022

ANSWER ALL QUESTIONS

INSTRUCTIONS

Select the correct ANSWER and **circle the letter corresponding to the correct answer in PEN**. For questions in part II involving calculations, you are required to show detailed working on the question paper to arrive at the answer you have selected.

Part I

1. When there is mass transfer across the system boundary, the system is called
   (a) isolated system
   (b) closed system
   (c) open system
   (d) none of the above

2. Steady flow means that the rates of flow of mass and energy across the control surface
   (a) varies
   (b) remains constant
   (c) depends on the control surface
   (d) none of the above

3. Which of the following represents the steady flow energy equation?
   (a) $Q - W = m\,[(h_2-h_1) + (V_2-V_1)\,(V_2-V_1)/2 + g\,(Z_2-Z_1)]$
   (b) $Q - W = m\,[(h_2-h_1) + (V_2-V_1)\,(V_2+V_1)/2 + g\,(Z_1-Z_2)]$
   (c) $Q - W = m\,[(h_1-h_2) + (V_2-V_1)\,(V_2+V_1)/2 + g\,(Z_2-Z_1)]$
   (d) $Q - W = m\,[(h_2-h_1) + (V_2-V_1)\,(V_2+V_1)/2 + g\,(Z_2-Z_1)]$

4. When more than one fluid stream enters or leaves the control volume, which type of balance is taken?
   (a) mass balance
   (b) energy balance
   (c) mass balance and energy balance
   (d) none of the above

5. What does a nozzle do?
   (a) decreases the velocity of a fluid at the cost of its pressure gain
   (b) increases the velocity of a fluid at the cost of its pressure drop
   (c) increases the velocity of a fluid and also its pressure
   (d) none of the above.

6. For a turbine, it is seen that work is done by the fluid at the expense of its enthalpy.
   (a) true
   (b) false

7. The limitation of the first law is
   (a) it does not indicate the direction of any spontaneous process
   (b) it assigns a quality to different forms of energy
   (c) it indicates the direction of any spontaneous process
   (d) none of the above

8. In a constant volume process, the internal energy change is equal to
   (a) heat transferred
   (b) work done
   (c) zero
   (d) none of the above

9. The change in enthalpy of a system is due to heat supplied at
   (a) constant volume
   (b) constant pressure
   (c) both at constant volume and pressure
   (d) none of the above

10. In the differential form, the steady flow energy equation becomes
    (a) $\delta q - \delta w = dh + VdV - gdZ$
    (b) $\delta q - \delta w = dh + VdV + gdZ$
    (c) $\delta q - \delta w = dh - VdV - gdZ$
    (d) $\delta q - \delta w = dh - VdV + gdZ$

11. Fluid flow through which of the following throttles the flow?
    (a) partially opened valve
    (b) orifice
    (c) porous plug
    (d) all of the above

12. Energy is a
    (a) point function
    (b) property of the system
    (c) extensive property
    (d) all of the above

13. If a fixed mass of steam is considered as the thermodynamic system, then the energy equation becomes
    (a) $Q = \Delta KE + \Delta PE - \Delta U + W$
    (b) $Q = \Delta KE + \Delta PE - \Delta U - W$
    (c) $Q = \Delta KE + \Delta PE + \Delta H + W$
    (d) $Q = \Delta KE + \Delta PE + \Delta U + W$

14. For an insulated nozzle, steady flow energy equation of the control surface gives (considering change in PE is zero and inlet velocity is small compared to exit velocity)
    (a) $V_2 = \text{sqrt}(4*\Delta h)$
    (b) $V_2 = \text{sqrt}(\Delta h)$
    (c) $V_2 = \text{sqrt}(\Delta h/2)$
    (d) $V_2 = \text{sqrt}(2*\Delta h)$

15. In an adiabatic compressor or pump,
    (a) the enthalpy of fluid remains constant with the amount of work input
    (b) the enthalpy of fluid decreases by the amount of work input
    (c) the enthalpy of fluid increases by the amount of work input
    (d) none of the above

16. A definite area or space where some thermodynamic process takes place is known as:
    (a) thermodynamic system
    (b) thermodynamic process
    (c) thermodynamic cycle
    (d) thermodynamic state

17. An open system is one in which
    (a) Heat and work cross the boundary of the system, but the mass of the working substance does not.
    (b) Mass of the working substance crosses the boundary of the system but heat and work do not.
    (c) Heat and work as well as mass of the working substance cross the boundary of the system.
    (d) Neither heat and work nor mass of the working substance cross the boundary of the system.

18. A given closed system undergoes a process in which work done by the system equals a decrease in the internal energy. The system must have undergone an
    (a) isothermal process
    (b) adiabatic process
    (c) isobaric process
    (d) isochoric process

19. Which thermodynamics law states that; two systems in thermal equilibrium with a third system individually are in thermal equilibrium with each other?
    (a) The third law
    (b) The second law
    (c) The zeroth law
    (d) The first law

20. Which of the following is an intensive property of a thermodynamic system?
    (a) Volume
    (b) Density
    (c) Mass
    (d) Energy

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Part II

21. A rigid tank contains a hot fluid that is cooled while being stirred by a paddle wheel. Initially, the internal energy of the fluid is 800 kJ. During the cooling process, the fluid loses 500 kJ of heat, and the paddle wheel does 100 kJ of work on the fluid. The final internal energy of the fluid is

B

(a) 200 kJ
(b) 400 kJ
(c) 1200 kJ
(d) 1400 kJ

$Q_{net} - W_{net} = \Delta U$

$-500 + 100 = U_2 - 800$

$U_2 = 400\ kJ$

3

Use the information provided below to answer questions 22 and 23

A quantity of fluid occupies 0.56 m^3 at 1 bar. After a compression which follows the law PV^n = constant, the pressure and volume are 5.3 bar and 0.18 m^3 respectively. If the internal energy of the fluid increases by 70 kJ during the process,

22. Calculate the value of the index, n

B

(a) 1.33
(b) 1.47
(c) 1.54
(d) 1.29

$1 \times (0.56)^n = 5.3 \times (0.18)^n$

$n = \ln(5.3/1) / \ln\left(\frac{0.56}{0.18}\right)$

$= 1.47$

3

23. Calculate the heat energy transfer

A

(a) 14 kJ
(b) 35 kJ
(c) 52 kJ
(d) 84 kJ

$W = \frac{P_2V_2 - P_1V_1}{1-n} = \frac{(100 \times 0.56) + (530 \times 0.18)}{1-1.47}$

$= -83.83\ kJ$

$Q = \Delta U + W$

$= 70 - 83.83 = -13.8\ kJ$

3

24. A fluid at 12.5 bar having a specific volume of 0.18 m^3/kg is throttled to a pressure of 5 bar. If the specific volume of the fluid after throttling is 0.101 m^3/kg, calculate the change in specific internal energy of fluid during the process.

B

(a) 198 kJ/kg
(b) 175 kJ/kg
(c) 157 kJ/kg
(d) 184 kJ/kg

$h_1 = h_2 \Rightarrow u_1 + P_1v_1 = u_2 + P_2v_2$

$u_2 - u_1 = P_1v_1 - P_2v_2$

$= (1250 \times 0.18) - (500 \times 0.101)$

$= 174.5\ kJ/kg$

3

Use the information provided below to answer questions 25 and 26

An insulated cylinder device containing 4.5 kg of water as shown in **Fig. Q25-26** expands at a constant pressure of 175 kPa. The initial volume and specific enthalpy of the water are 0.5 m^3 and 485 kJ/kg, respectively. The water is stirred by a paddle wheel amounting to a work transfer of 400 kJ while a current of 8 A flows for 45 mins through a resistor placed in the water. If the final specific enthalpy is 1594 kJ/kg;

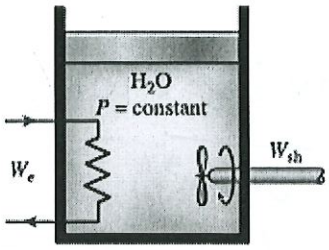


Fig. Q25-26

$U_1 = H_1 - P_1V_1$ (3)

$= m(h_1 - P_1v_1) = 4.5 \times (485 - (175 \times \frac{0.5m^3}{4.5kg}))$

$= 2095kJ$

25. Determine the initial internal energy of the water
    (a) 108.5 kJ
    (b) 465.6 kJ
    (c) 397.5 kJ
    (d) 2182.5 kJ

2095kJ

$Q_{net} - W_{net} = \Delta U$

$-(W_b - W_e - W_{sh}) = \Delta U$

26. Determine the voltage of the source
    (a) 241.7 V
    (b) 270.6 V
    (c) 214.7 V
    (d) 212.5 V

D

$400kJ + \frac{(8 \times 45 \times 60)V}{1000} = 4.5(1594 - 485)kJ$

$V = 212.5V$ (3)

Use the information provided below to answer questions 27 to 30

A nozzle as shown in **Fig. Q27-30** is supplied with steam having a specific enthalpy of 3268 kJ/kg, specific volume of 0.4 m^3/kg and velocity of 10 m/s. At the exit of the nozzle, the specific volume and specific enthalpy of the steam are 1.32 m^3/kg and 3072 kJ/kg, respectively while losing heat at a rate of 25 kW. For an inlet area of 800 cm^2,

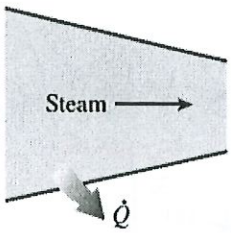


Fig. Q27-30

27. Determine the mass flow rate
(a) 1.5 kg/s
(b) 2.0 kg/s
(c) 2.5 kg/s
(d) 3.0 kg/s

B

$\dot{m} = \frac{A_1 C_1}{v_1} = \frac{800 \times 10^{-4}\ m^2 \times 10\ m/s}{0.4\ m^3/kg}$

$= 2.0\ kg/s$

28. Determine the exit velocity
(a) 303 m/s
(b) 540 m/s
(c) 606 m/s
(d) 715 m/s

C

626.18 m/s

$h_1 + \frac{1}{2}C_1^2 = h_2 + \frac{1}{2}C_2^2$

$3268 + \frac{10^2}{2000} = 3072 + \frac{C_2^2}{2000} + 12.5\ kJ/kg$

$C_2 = 605.85\ m/s = 606\ m/s$

29. Determine the change in kinetic energy
(a) 150 kJ/kg
(b) 184 kJ/kg
(c) 205 kJ/kg
(d) 248 kJ/kg

B

$\Delta KE = \frac{1}{2}(C_2^2 - C_1^2) = \frac{606.18^2 - 10^2}{2000}$

$= 196\ kJ/kg$ 183.571 kJ/kg.

30. Determine the exit volume flow rate
(a) 1.85 m^3/s
(b) 2.45 m^3/s
(c) 2.64 m^3/s
(d) 3.21 m^3/s

C

$\dot{m} = \frac{A_2 C_2}{v_2}$

$\dot{V} = \dot{m} v_2 = 2.0\ kg/s \times 1.32\ m^3/kg$

$= 2.64\ m^3/s$

K. O. Amoabeng